Chapter 2 Geological time

Extra worksheet 1 answers: Carbon-14 dating

1 A carbon containing compound produced by a living thing.

2 Carbon-14 is formed high in the atmosphere by cosmic rays (radiation from the Sun) changing nitrogen atoms into carbon-14 atoms.

3 The radioactive carbon dioxide is absorbed by plants in photosynthesis and made into sugar and later into other organic compounds in the plant.

4 When the plant is eaten by animals, the radioactive carbon atoms become part of the animal's body tissues.

5 5730 years old.

6 About 11 460 years old.

7 Rock cannot be dated using carbon-14 dating unless it contains material that was originally part of a living organism (such as coal). Few rocks are of organic origin or contain any organic materials such as oil.

8 It could not be used because the method is only accurate up to about 50 000 years.

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